Bihar Engineering University, Patna

B.Tech 3rd Semester Examination, 2024

Course: B.Tech | *Time: 03 Hours*
Code: 100306 | *Subject: Electrical Circuit Analysis* | *Full Marks: 70*

Instructions: -

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
*(iv) Question No. **1** is compulsory.*

Q.1 ***Choose the correct answer from the following (Any seven question only):*** **[2 x 7 = 14]**

(a) Thevenin's and Norton's equivalents are:
(i) Different representations with different values
(ii) Identical in all circuits
(iii) Equivalent with same behavior at load terminals
(iv) Applicable only for AC circuits

(b) Mesh analysis is preferred when:
(i) The number of meshes is more than nodes
(ii) The number of meshes is less than nodes
(iii) Dependent sources are present
(iv) Circuits are complex

(c) The steady-state response in a circuit is observed:
(i) During initial t = 0
(ii) When $t \to \infty$
(iii) When energy is maximum
(iv) Just after switching

(d) In a critically damped RLC circuit, the roots of characteristic equation are:
(i) Complex conjugates
(ii) Real and distinct
(iii) Real and equal
(iv) Purely imaginary

(e) The admittance of an inductor is:
(i) $j\omega L$
(ii) $1/(j\omega L)$
(iii) $-j\omega L$
(iv) $j\omega C$

(f) The average power in a purely capacitive circuit is:
(i) Zero
(ii) Maximum
(iii) Depends on frequency
(iv) Infinite

(g) Poles are values of 's' where:
(i) Input is zero
(ii) Output is zero
(iii) Denominator of transfer function is zero
(iv) Numerator is zero

(h) Convolution in time domain becomes.....in Laplace domain:
(i) Division
(ii) Addition
(iii) Multiplication
(iv) Integration

(i) A network is reciprocal if it contains only:
(i) Resistors
(ii) Resistors and capacitors
(iii) Passive linear elements
(iv) Active devices

(j) The h-parameters are mainly used for:
(i) Filter design
(ii) Inductor modeling
(iii) Transistor modeling
(iv) Transformer design

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Q.2 (a) State Maximum Power Transfer Theorem. Prove that efficiency of the circuit under maximum power transfer condition is 50%. [7]

(b) Find the voltage across 3Ω resistor using any theorem for the circuit shown in Figure-1 [7]

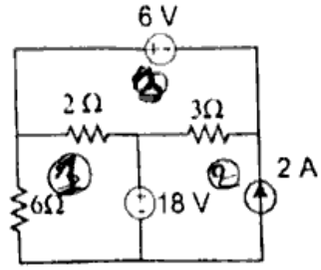


Figure 1

Q.3 (a) Find V_A using the principle of superposition in network as shown in Figure 2. [7]

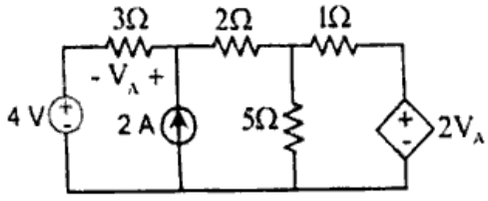


Figure 2

(b) State Reciprocity Theorem. Verify the Reciprocity Theorem for the circuit shown in Figure 3. [7]

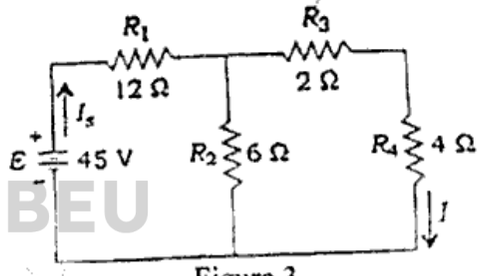


Figure 3

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Q.4 (a) Explain the behaviour of RL and RC elements for transient condition. Mention their representation at the time of switching. [7]

(b) Draw the circuit of parallel RLC circuit and derive the formulae for resonant frequency, half-power frequencies, bandwidth and quality factor. [7]

Q.5 (a) Determine Laplace transform of a following (i) sin2t (ii) cos2t [7]

(b) Synthesis the waveform shown in Figure 4 below and Determine Laplace transform of periodic waveform. [7]

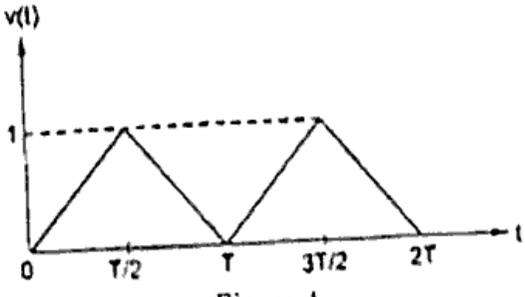


Figure 4

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Q.6 (a) To find Thevenin equivalent between terminals 'a' and 'b' as shown in Figure 5. [7]

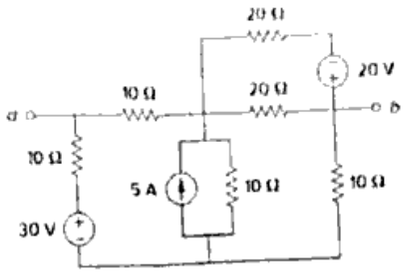


Figure 5

(b) A switch is moved from position 1 to 2 at t=0 as shown in Figure 6. Find $V_R(t)$ and $V_C(t)$ for t>0. [7]

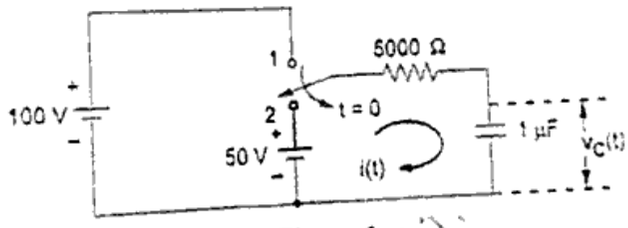


Figure 6

Q.7 (a) Explain about the transmission parameters and derive the condition for symmetry and reciprocity. [7]

(b) Determine z-parameter of network shown below in Figure 7 & give its equivalent circuit. [7]

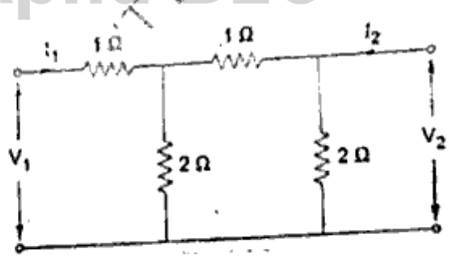


Figure 7

Q.8 (a) Define two-port network and write assumptions to be made to find network and also obtain Y parameter. [7]

(b) Two coils connected in series have an equivalent inductance of 10H. When the connections of one coil are reversed, the effective inductance is 6H. If the co-efficient of coupling is 0.6, calculate the self-inductance of each coil and mutual inductance. [7]

[7 x 2 = 14]

Q.9 Write short notes on *any two* of the following:

(a) Poles and Zeros
(b) Convolution integral
(c) Mutual coupled circuits
(d) Transmission and Hybrid parameters

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